

Variability and acceleration in the growth of children's early vocabulary

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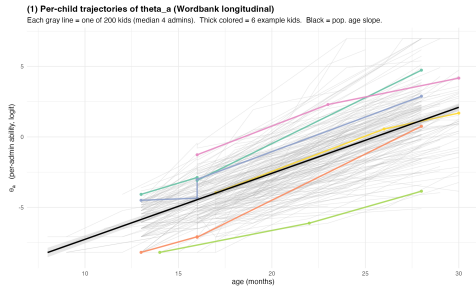
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Vocabulary growth: rapid, variable, consequential

Theoretical: Growth mechanisms are diagnostic of how children's learning works — and how it changes with development.

Practical: Variation in vocabulary growth is a key precursor of academic outcomes.

Methodological: Vocabulary count has a property nearly no other psychological metric has — a *true zero*. Children at birth know zero words. Absolute variability in word counts is interpretable, not just z-scores.



Per-admin $\theta_{i,a}$ on Wordbank longitudinal sample. Each grey line = one child.

Two signatures of vocabulary growth

What any theory of word learning has to predict

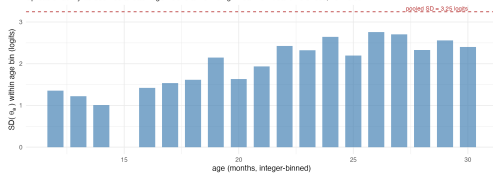
Variability

Children fan out, not converge.

Within-age SD of ability *grows* with age.

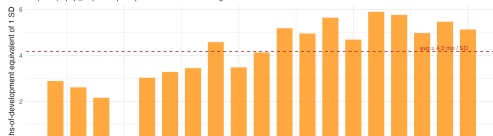
(2a) Within-age SD of theta_a vs. age

Spread of ability across kids at each age. Constant or rising = real individual differences, not noise.



(2b) Translation: 1 SD across kids = how many months of dev?

SD(theta) / pop_slope. Pop. slope estimated as 0.468 logits/mo.



Acceleration

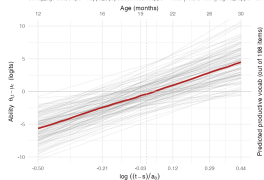
Vocabulary grows super-linearly in tokens.

Population scaling exponent $\kappa_{\text{pop}} \gg 1$.

M best (long no freq slopes): per-child trajectory fan

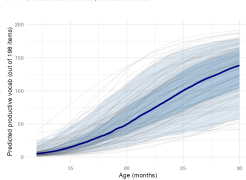
Per-child ability trajectories

120 kids sampled from posterior population MVM; red = population median.
On log_age axis, slope = kappa_1; population mean kappa_1 = 10.35 [10.21, 10.51]; sigma_kappa_1 = 3.47



Same kids, vocabulary scale (super-linear in age)

Voiced = $\text{sem}_i \cdot \text{sigmoid}(\theta_{\text{full}} + \log_H + \text{ps}_i)$ across the 198 CDI items.
Blue ribbons: 25/75 and 5/95 percentiles across kids.



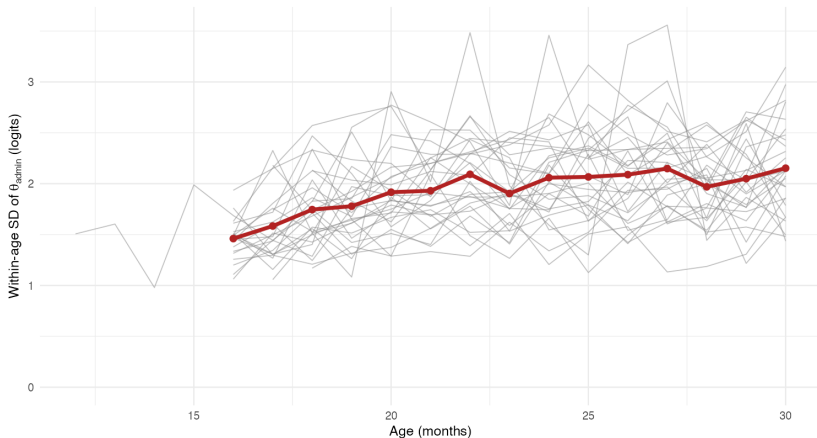
Cross-language replication

Variability and growth-with-age across Wordbank languages

Cross-language replication (N = 29 languages)

Each grey line = one language; red = cross-language mean.

Within-age variability in productive vocabulary is a universal pattern, growing with age.



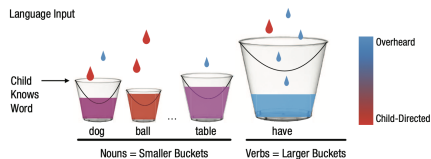
The structural null: pure accumulator

Where prior models sit, and what they all lack

Pure accumulator. Each child accumulates word-evidence at per-child input rate r_i ; word j is acquired when threshold ψ_j is crossed.

Per-child scaling exponent κ_i : ability scales as (cumulative tokens) $^{\kappa_i}$.

- $\kappa_i = 1$ for every child: pure unit accumulator.
- Between-child variance is determined entirely by σ_r .
- No per-child slope variation around the population mean.



Bucket-fill metaphor: each child hears tokens at rate $r_i \cdot p_j$; word j is acquired when its bucket fills past threshold ψ_j (Kachergis et al. 2021).

Two estimands diagnose deviation from the null:

Where prior accumulator models sit

None has both per-child efficiency and per-child slope variation

Model	per-child σ_α	per-child σ_κ	$\kappa_{\text{pop}} > 1$
McMurray (2007) – pure accumulator	–	–	only via $g'(\tau)$
Mitchell & McMurray (2009) – + leverage	–	–	shifts but cannot create
Hidaka (2013) – closed-form AoA	–	–	via rate-change exponent
Mollica & Piantadosi (2017) – Gamma	–	–	embedded in per-word
Kachergis et al. (2021) – 1-PL Rasch	θ_i free	–	linear age covariate
M_best (this work)	free	free	free

Common gap. *No prior accumulator model has both σ_α and σ_κ free.* Most have neither. The two signatures we measure — variability and acceleration in growth rate — are precisely what these models cannot produce.

Implication for the next slides. The 4-panel comparison shows each model's predictions for the canonical CDI percentile fan. The nested family makes clear what each addition buys.

Linear predictor

Item-response theory + accumulator dynamics

$$P(\text{produces}_{ijt} = 1) = \sigma(\eta_{ijt})$$
$$\eta_{ijt} = \underbrace{\xi_i}_{\text{child}} + \underbrace{\kappa_i}_{\text{scaling exponent}} \log \frac{t}{a_0} - \underbrace{\psi_j}_{\text{word}}$$

Child intercept $\xi_i = \log r_i + \log \alpha_i$.

- $\log r_i$: input rate, externally pinned via Sperry / HR / WF.
- $\log \alpha_i$: efficiency multiplier (free).

Scaling exponent $\kappa_i \sim \mathcal{N}(\kappa_{\text{pop}}, \sigma_{\kappa}^2)$: ability scales as $(\text{tokens})^{\kappa_i}$. McMurray's pure accumulator sits at $\kappa_i = 1$ for every child.

Word threshold ψ_j : log-tokens needed to acquire word j , hierarchical by lexical class.

Headline derived quantities:

- $\pi_{\alpha} = \sigma_{\alpha}^2 / (\sigma_{\alpha}^2 + \sigma_r^2)$: efficiency-share of intercept variance.
- $\kappa_{\text{pop}}, \sigma_{\kappa}$: population scaling and between-child variation in scaling.

M_best fit: per-child trajectory fan

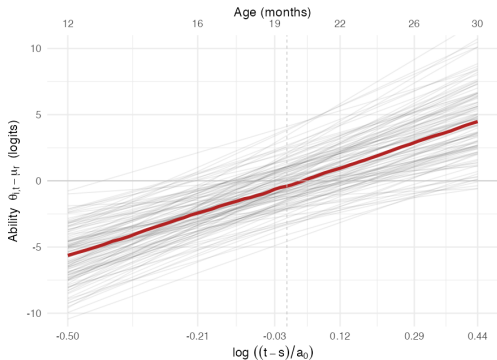
120 kids drawn from posterior of `long_no_freq_slopes`

M_best (long_no_freq_slopes): per-child trajectory fan

Per-child ability trajectories

120 kids sampled from posterior population MVN; red = population median.

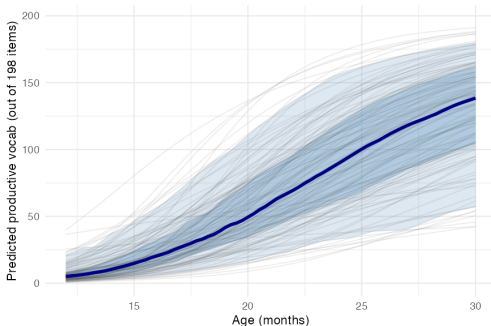
On `log_age` axis, slope = `kappa_i`; population mean `kappa` = 10.35 [10.21, 10.51]; `sigma_kappa` = 3.47.



Same kids, vocabulary scale (super-linear in age)

Vocab = $\sum_j \text{sigmoid}(\theta_{it,j} + \log_H - \psi_{i,j})$ across the 198 CDI items.

Blue ribbons: 25/75 and 5/95 percentiles across kids.



`psi_j` source for vocab calibration: M_best-equivalent (`long_slopes - log_p`). Left panel x-axis is $\log((t-s)/a_0)$ so the M_best slope ($1+\delta+\zeta+\eta$) appears linear; right panel x-axis is age in months for direct vocab interpretation.

Left: ability $\theta_{it} - \mu_r$ vs. $\log_{age}$, slope = κ_i , mean ≈ 10.4 . **Right:** same kids, vocabulary on ^{8 / 19}

M_best parameters

Variability and acceleration both bounded firmly above zero

Parameter	Posterior	What it does
σ_α	1.71 [1.55, 1.91]	Between-child efficiency variation.
π_α	0.91 [0.89, 0.93]	Share of intercept variance from efficiency, not input.
κ_{pop}	10.35 [10.21, 10.51]	Population scaling exponent: ability scales as (tokens)$^\kappa$.
σ_κ	3.47 [3.14, 3.84]	Between-child variation in scaling exponent.
$\rho(\xi, \kappa)$	+0.35 [+0.22, +0.47]	Higher-baseline kids have steeper growth.

Variability

$\pi_\alpha = 0.91$: efficiency variance dominates input variance.

Acceleration

$\kappa_{\text{pop}} \approx 10.4 \gg 1$: scaling is wildly super-linear, far from the unit accumulator.

Variation in scaling

$\sigma_\kappa > \sigma_\alpha$: kids vary more in scaling exponent than in baseline.

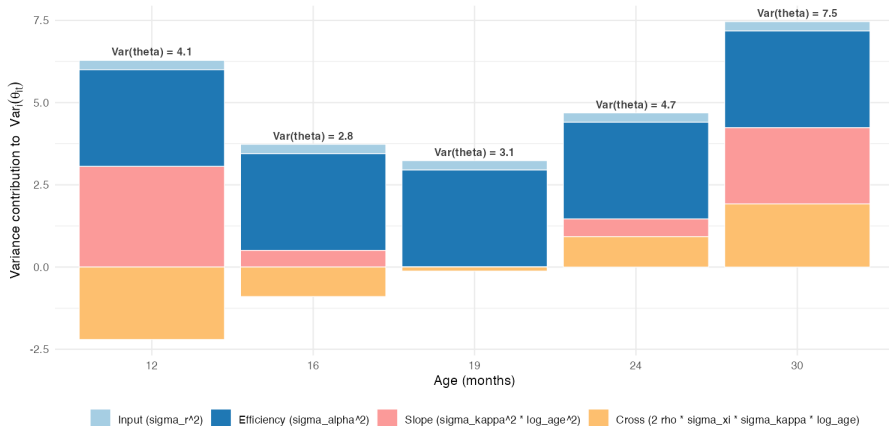
Where between-child variance comes from, by age

π_α is exact at a_0 and qualified elsewhere

Between-child variance composition by age (M_best)

$\sigma_r=0.53$ (input), $\sigma_\alpha=1.71$ (efficiency), $\sigma_\kappa=3.47$ (slope spread), $\rho(x,\kappa)=0.35$.

At $a_0=19$ mo, $\log_age=0 \rightarrow$ slope and cross terms vanish; $\pi_\alpha = \sigma_\alpha^2 / (\sigma_\alpha^2 + \sigma_r^2) = 0.91$ exact.



Cross-sample variability: π_α replicates

Five samples, two languages, three input-observation channels

Sample	N	σ_α	π_α
English Wordbank longitudinal	200	1.83 [1.65, 2.04]	0.91 [0.90, 0.93]
Norwegian Wordbank longitudinal	200	2.10 [1.90, 2.35]	0.94 [0.93, 0.95]
BabyView (head-mounted video)	20	1.13 [0.86, 1.61]	0.84 [0.68, 0.93]
SEEDLingS (LENA + comp correction)	44	1.39 [1.13, 1.74]	0.94 [0.89, 0.97]
Peekbank-Stanford (LWL processing)	62	1.64 [1.34, 2.03]	0.90 [0.86, 0.94]

Headline. $\pi_\alpha \in [0.84, 0.94]$ across every well-conditioned sample.

Input rate quantity explains <10% of between-child intercept variance.

Cross-channel replication: directly observed input (BabyView, SEEDLingS), inferred input (Wordbank), processing-coupled (Peekbank). SEEDLingS extreme corrected for parent-report noise via comprehension channel.

Cross-language acceleration: where it lives, not whether it does

Sample	admins/kid	κ_{pop}	σ_{κ}	kid range ($\kappa_{\text{pop}} \pm 2\sigma_{\kappa}$)
English	median 3	10.4 [10.2, 10.5]	3.5 [3.1, 3.8]	~ 3 to 17
Norwegian	median 8	12.4 [12.3, 12.6]	3.7 [3.4, 4.2]	~ 5 to 20

Population scaling exponent $\kappa_{\text{pop}} \sim 10-12$ in both languages. Implied $P(\kappa_i > 1) > 0.997$ in both.

Internal decomposition into δ vs. ζ_i is

identifiability-dependent on longitudinal density.

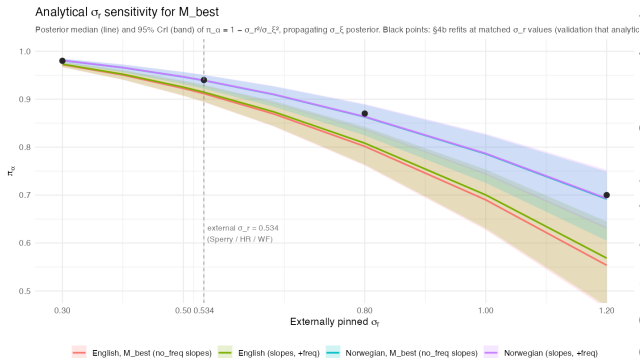
In Norwegian (8 admins/kid) per-child slopes are well-identified, and the population term δ alone — without per-child deviations — hurts fit by 235 ELPD. In English (3 admins/kid), δ pools what Norwegian's data lets fall into ζ .

Reading. The acceleration signature is universal. The parameterization that captures it is identifiability-dependent on data density.

Concretely: in any sufficiently dense longitudinal sample, between-child κ_i spans roughly 5–15.

Stress test 1: how robust is π_α to the input-rate prior?

Analytical sensitivity, validated by refits



$$\pi_\alpha(\sigma_r) = 1 - \sigma_r^2/\sigma_\xi^2.$$

The data identifies σ_ξ^2 tightly; the σ_r prior just *partitions* it into input vs. efficiency.

At external $\sigma_r = 0.534$:

$\pi_\alpha \approx 0.91 - 0.94$.

At doubled $\sigma_r = 1.2$: π_α still > 0.55 .

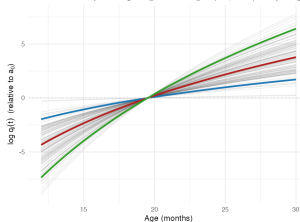
Conclusion. Efficiency-dominated decomposition holds across any defensible external σ_r .

Stress test 2: can input quality absorb acceleration?

Quality-multiplier interpretation: same model, different label

Implied per-child quality multipliers $q_i(t)$

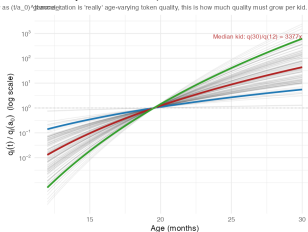
Under the structural equivalence: $\gamma_{\text{token}} = \delta + \alpha \cdot \text{age}_i \sim N(9.35, 3.47^2)$. Quality must grow as $(t/a_0)^{\gamma_{\text{token}}}$ to absorb M_best's acceleration. This is how much quality must grow per kid.



kid percentile — 5% — 50% — 95%

Posterior of M_{best} gives $\gamma_{\text{token}} = \delta + \alpha \cdot \text{age}_i \sim N(9.35, 3.47^2)$. For "token quality grows with age" to absorb M_{best} 's acceleration, median kid needs $q(30)/q(12) \sim 3377\times$, ± 1 SD spread covers 40x to 26000x.

Same trajectories on multiplicative scale



kid percentile — 5% — 50% — 95%

Reviewer reflex. “Maybe kids’ tokens differ in *quality*, and that’s what looks like acceleration.”

Structural answer. Per-child age-varying quality $q_i(t) = (t/a_0)^{\gamma_i}$ on input is observationally equivalent to $\gamma_i = \kappa_i - 1$.

Implication. Quality cannot escape acceleration — it *relabels* it. For the median kid, “quality” would have to grow $\sim 3,000\times$ from age 12 to 30 mo. For the 95th percentile kid, $\sim 10^6\times$. Implausible as a substantive quality story.

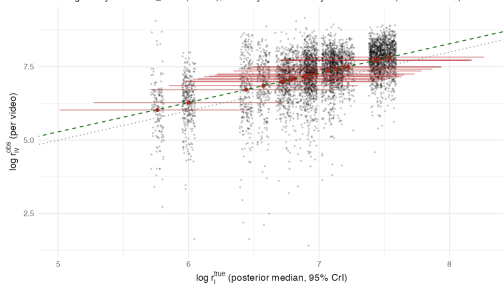
Replicating with directly observed input

BabyView (head-mounted video) and SEEDLingS (LENA all-day audio)

BabyView, $N = 20$

Per-child observed input vs inferred true rate

Dashed green: $y = x + \text{beta_react}$ ($= 0.28$); dotted: $y = x$. Reactivity inflation = 33% (CrI: -35-173%)

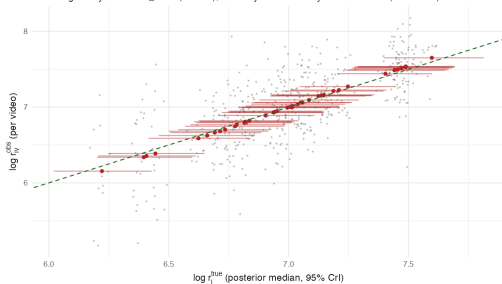


$$\pi_{\alpha} = 0.84 [0.68, 0.93]$$

SEEDLingS, $N = 44$ (with comp correction)

Per-child observed input vs inferred true rate

Dashed green: $y = x + \text{beta_react}$ ($= -0.00$); dotted: $y = x$. Reactivity inflation = -0% (CrI: -0-0%)



$$\pi_{\alpha} = 0.94 [0.89, 0.97]$$

With per-recording observed input rates, the π_{α} headline holds. Input-rate variance is a small

Adding a processing-speed readout: Peekbank LWL channel

$N = 62$ Stanford-linked subjects with both CDI and looking-while-listening

Joint model adds a per-child LWL log-RT readout coupled to $\log \alpha_i$ via γ_{rt} .

Posterior.

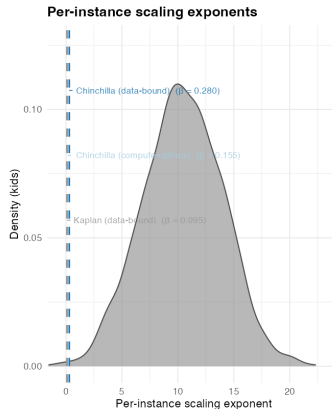
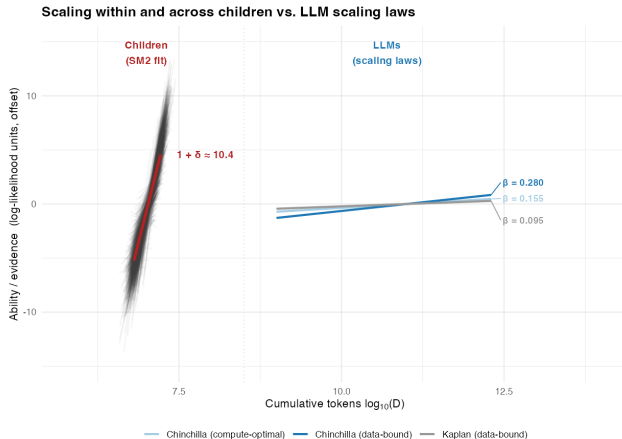
- $\gamma_{rt} = 0.082$ [0.046, 0.125] — LWL processing speed is real, and bounded above zero.
- $\pi_{\alpha} = 0.90$ [0.86, 0.94] — replicates within the Peekbank subset.
- $\rho(\kappa_i, \text{rtslope}_i) = -0.02$ [-0.33, +0.27] — CDI-side and LWL-side growth rates do *not* share variance.

Two clocks, not one. The acceleration in vocab growth and the maturation of processing speed are independent maturation processes within the same child — different “efficiency clocks” ticking in parallel.

This rules out the simplest unification (“one underlying efficiency axis”); the two readouts each pick up a different developmental process.

Coda: how does this compare to LLM scaling laws?

Slopes are the structurally meaningful comparison; y-axis offset is anchored for visual clarity. Kid D range comes from per-kid $r_{\perp}$.



SM2 fit 'long_slopes': median $1+\delta = 10.39$ [10.24, 10.55], $\sigma_{\zeta} = 3.48$ [3.15, 3.87], $\mu_{\zeta} = 7.34$, $\sigma_{\mu_{\zeta}} = 0.53$, $a_{\mu} = 19$. Chinchilla and Kaplan exponents from Hoffmann et al. 2022 / Kaplan et al. 2020.

Mechanisms not in the standard model

What human learning needs that pure accumulators don't have

Sources of variability

- Interactional input quality (joint attention, contingent talk)
- Genetic / heritable variation
- Environmental richness not captured by token quantity
- Processing-side individual differences (working memory, attention)

Sources of acceleration

- Growing processing efficiency with maturation
- Learning-to-learn (mutual exclusivity, shape bias, syntactic bootstrapping)
- Domain-general changes in attention, memory, perception
- Cross-word leverage / vocabulary-mediated bootstrapping

The structural finding constrains theory: any account of early word learning must produce both super-linear-in-tokens growth and large per-child variation in growth rate.

Summary

- Vocabulary uniquely affords *absolute* variability measurement — there is a true zero, and word counts are interpretable in word-count units.
- Two robust signatures across ~ 30 languages and 5 modeling samples: **variability** ($\pi_\alpha \approx 0.91$, growing with age) and **acceleration** ($1 + \delta + \zeta_i \sim 10$, varying widely across kids).
- Both signatures violate a pure-accumulator null. Posited “quality” alternatives *relabel* acceleration rather than escape it.
- The picture is structurally different from LLM scaling laws on both axes (steeper exponent, between-instance variance).
- Open question: *which* mechanism — maturation, learning-to-learn, processing speed — is the load-bearing source of the acceleration we measure?